25901714 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCIAM101 - AUTOMOTIVE SYSTEMS FOR E - VEHICLES

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. In a non-inverting amplifier, if the feedback resistor (Rf) is increased, the gain of the amplifier will: [K] / [CO1]

   a). Decrease

   b). Become negative

   c). Remain unchanged

   d). Increase

2. The primary function of a bridge rectifier is [U] / [CO2]

   a). To amplify an AC signal

   b). To convert AC voltage into DC voltage

   c). To filter the DC voltage

   d). To stabilize the DC output voltage

3. What is the main challenge of wireless charging for EVs? [U] / [CO3]

   a). High energy efficiency

   b). Low compatibility with modern EVs

c). Energy losses during transmission

d). Overheating of the battery during charging

4. Which of the following systems uses an open-loop configuration? [U] / [CO1]

a). A system with feedback

b). A simple amplifier without feedback

c). A system with a control loop

d). A system that adjusts output automatically

5. How is the rotation of a BLDC motor typically controlled? [A] / [CO3]

a). By varying the supply voltage to the rotor

b). By using an external commutator

c). By switching the current in the stator windings using an electronic controller

d). By varying the magnetic field strength of the rotor

6. In sensor applications, the role of operational amplifier is [U] / [CO1]

a). Signal filtering

b). Signal amplification

c). Signal conversion

d). Data storage

7. What happens if the feedback loop of SMPS fails? [U] / [CO2]

a). The output voltage becomes unstable

b). The switching frequency increases

c). The system becomes more efficient

d). The transformer gets damaged

8. What happens when an ESC detects that the motor has reached the desired speed? [A] / [CO3]

a). It disconnects the battery

b). It reduces the power to maintain the speed

c). It increases the voltage to the motor

d). It shuts off the motor

9. What happens when the input voltage exceeds a certain threshold in a Voltage Level Indicator circuit using a 741 op-amp? [U] / [CO2]

a). The LED corresponding to that threshold voltage turns on

b). The output of the op-amp becomes zero

c). The reference voltage is adjusted automatically

d). The input voltage is reduced to the reference level

10. At what state of charge (SOC) does DC fast charging typically slow down? [K] / [CO3]

a). 10% SOC

b). 50% SOC

c). 80% SOC

d). 100% SOC

[1x10 = 10]

Part B

Short Answer Questions

Answer any **4** questions

Each question carries **5** marks

11. Explain the basic functionality of an Operational Amplifier and list the ideal characteristics of an Op-Amp. [U] / [CO1]

12. Describe the charging and discharging process of a lithium-ion battery [U] / [CO2]

13. How does a capacitor filter work in a bridge rectifier circuit? [U] / [CO2]

14. Explain pin configuration of LM324. [U] / [CO1]

15. How can we use a comparator as zero crossing detector [U] / [CO2]

16. Explain how the TP4056 ensures safe charging of lithium-ion batteries. [K] / [CO3]

[5x4 = 20]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

17. What is an LDR. Expain how we can use an op-amp to amplify the signal from a LDR with proper circuit diagram [K] / [CO1]

18. Explain the working of Induction motor [U] / [CO3]

19. How do input and output capacitors help improve the performance of a 78XX voltage regulator? [U] / [CO2]

20. Briefly explain unity gain amplifier with proper circuit diagram. What is the gain of a unity gain amplifier? [K] / [CO1]

[10x2 = 20]